

# PHYSICAL AI CHALLENGE

Robotic defect detection, VLA reasoning, and optimized on-device execution



## Challenge Statement

Build an end-to-end Physical AI solution in which an SO-101 robotic arm performs autonomous defect detection and the appropriate physical response. Teams will use Intel Physical AI Studio to design the Vision-Language-Action (VLA) workflow, Anomalib for visual anomaly and defect detection, and OpenVINO to optimize and deploy the implemented AI pipeline on an Intel Core Ultra Series 3 processor using the integrated NPU, GPU, and CPU where appropriate.

## Challenge Description

Build an end-to-end Physical AI solution for robotic defect detection and intelligent manipulation using the Intel Physical AI Studio, Anomalib, and OpenVINO software stack. Participants will design a robotics workflow that combines Vision-Language-Action (VLA) capabilities through Intel Physical AI Studio with Anomalib-based visual anomaly and defect detection. The solution should enable the robot to perceive objects, identify defects, reason about the required action, and execute the appropriate manipulation task.

The completed solution will then be optimized and deployed on Intel Core Ultra Series 3 using OpenVINO, leveraging available CPU, GPU, and NPU acceleration to improve inference performance, latency, and deployment efficiency. The specific defect-detection scenario, objects, operating conditions, and task constraints will be presented at the start of Challenge Day.

## Technology Stack



## Challenge Objective

- Integrate Anomalib defect detection into a robotics application rather than treating vision inference as a standalone demo.
- Use Intel Physical AI Studio to connect perception outputs with VLA-based reasoning, task planning, and robot action.
- Optimize supported AI workloads with OpenVINO and demonstrate efficient execution on Intel Core Ultra Series 3 using NPU, GPU, and CPU resources where appropriate.
- Demonstrate a complete closed-loop workflow on the SO-101 robotic arm using the use-case details revealed at Challenge Day.

Challenge-Day reveal: the exact defect scenario, objects, task rules, and operating conditions will be provided at kickoff.

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## Required Technology Stack

Teams will build the solution using the following robotics and AI components as one integrated system.

### 01 SO-101 Robotic Arm



Physical manipulation platform used to execute the challenge task. The arm closes the loop from AI perception and reasoning to real-world action.

### 02 Intel Physical AI Studio

Design and integrate the VLA-based robotics workflow, connecting perception, task reasoning, robot control, and application orchestration.

**VLA WORKFLOW**

### 03 Anomalib

Provide visual anomaly and defect detection, producing defect decisions and localization signals that can be consumed by the VLA / robot workflow.

**DEFECT DETECTION**

### 04 OpenVINO + Intel Core Ultra Series 3

Optimize and deploy supported AI workloads with OpenVINO. Map inference across the integrated NPU, GPU, and CPU to improve latency and on-device efficiency.

**OPTIMIZE + DEPLOY**

## End-to-End Technical Workflow

The solution is a closed-loop robotics pipeline: perception identifies defects, VLA reasoning determines the required response, the robot executes the action, and OpenVINO optimizes deployment for Intel on-device compute.



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## Technical Implementation Expectations

### 1. Perception & Defect Detection

Capture object imagery from the robot workspace and apply Anomalib to identify anomalous or defective objects. The defect output should be available to the downstream robotics policy as a structured signal, label, score, or localized region.

### 2. VLA Reasoning & Task Planning

Use Intel Physical AI Studio to construct the VLA-driven application flow. The VLA layer interprets the task instruction and perception results, determines the appropriate response, and produces the action intent required by the robot controller.

### 3. SO-101 Robot Execution

Convert the selected action into executable robot behavior on the SO-101 arm. The implementation should connect AI decisions to robot motion, grasping, sorting, or other challenge-specific manipulation actions.

### 4. OpenVINO Optimization

Optimize supported models and inference stages with OpenVINO. Teams should consider model conversion, precision, graph optimization, preprocessing / postprocessing efficiency, and device selection to reduce end-to-end latency.

### 5. Intel Core Ultra Series 3 Deployment

Deploy the optimized solution on Intel Core Ultra Series 3. Use the integrated NPU, GPU, and CPU according to workload characteristics and operator support, while maintaining reliable robot control and application responsiveness.

### 6. Closed-Loop Demonstration

Demonstrate the full sequence from image acquisition and defect detection through VLA reasoning and physical robot action, followed by a concise explanation of the architecture and optimization decisions.

### Challenge-Day Format

The use case details will be presented at Challenge Day. Teams will receive the defect definition, task rules, target objects, and operating conditions at kickoff, then design and implement the solution using the required Intel robotics stack. Core requirement: the final system must connect defect detection to an autonomous robot action rather than demonstrate perception as an isolated model.

### What Teams Must Deliver

- Working Physical AI prototype: closed-loop SO-101 solution from perception to action.
- VLA + Anomalib integration: VLA robotics workflow with visual anomaly / defect detection.
- OpenVINO-optimized deployment: optimized inference pipeline on Intel Core Ultra Series 3.
- Live demonstration: execute the Challenge-Day task and explain the architecture and optimization choices.

## Engineering Focus Areas

Perception robustness: stable defect detection under the challenge setup. System latency: minimize perception-to-action delay. Heterogeneous compute: use NPU/GPU/CPU effectively through OpenVINO. Robot reliability: repeatable execution and recovery from incorrect or uncertain detections.

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## Judging Criteria

Solutions will be evaluated on technical integration, robotics execution, and efficient Intel deployment.

| Criteria                                          | Description                                                                                                                                                                                                                                                               | Points (100) |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| End-to-End Physical AI Solution                   | Demonstrates a complete workflow from visual perception and defect detection through decision-making and robotic action using the SO-101 arm. The solution should operate as one integrated closed-loop system rather than as independent component demonstrations.       | 25           |
| Defect Detection with Anomalib                    | Effective use of Anomalib for visual anomaly and defect detection. Evaluation considers detection quality, model integration, handling of normal versus defective objects, and how defect outputs are exposed to and consumed by the downstream robotics workflow.        | 20           |
| VLA & Physical AI Studio Integration              | Demonstrates effective use of Intel Physical AI Studio and VLA-based reasoning to translate perception results and task instructions into appropriate robot actions. Integration across perception, reasoning, planning, and manipulation should be clearly demonstrated. | 20           |
| OpenVINO & Intel Core Ultra Series 3 Optimization | Optimizes and deploys supported AI workloads with OpenVINO on Intel Core Ultra Series 3. Evaluation considers inference latency, throughput, deployment efficiency, and appropriate use of the integrated NPU, GPU, and CPU for the implemented workload.                 | 20           |
| Robotic Execution & Reliability                   | Quality and robustness of SO-101 execution, including object localization, manipulation accuracy, repeatability, stable task completion, and recovery from incorrect or uncertain defect detections during the Challenge Day demonstration.                               | 10           |
| Innovation & Technical Demonstration              | Originality of the approach, effective use of Intel technologies, overall technical completeness, and clarity of the live demonstration, system architecture, and optimization decisions.                                                                                 | 5            |
| Total                                             | Overall Challenge Score                                                                                                                                                                                                                                                   | 100          |

**Evaluation emphasis:** Integrated perception-to-action workflow and efficient deployment on Intel hardware. Teams should be prepared to explain architecture, workload placement, optimization choices, and observed results.